HINTS & SOLUTION

Q.1 (B)

Let the mass of sample = 100 g

Mass of nitrogen in the sample = 42 g = $\frac{42}{14}$ = 3 mole

$\therefore$ Mass of pure urea = $\frac{3}{2}\times 60$ = 90 g $\therefore$ Purity = 90%

$$\left[\begin{array}{ccc} \because & \frac{\text{Urea}}{1\text{mole}} & \frac{\text{N}}{2\text{ mole}} \\ & \frac{3}{2}\text{mole} \leftarrow & 3\text{ mole} \end{array}\right]$$

Q.2 (D)

$C_2H_6O\,(g) + 3O_2\,(g) \longrightarrow 2CO_2\,(g) + 3H_2O\,(l)$

20	80	0
0	80–60	40

$V_i = 20 + 80 = 100 \qquad V_f = 20 + 40 = 60$

reduction in volume = $V_i - V_f = 100 - 60 = 40$

% Reduction in volume = $\frac{40}{100}\times 100 = 40\%$

Q.3 (A)

$2MnO + 5PbO_2 \longrightarrow 2HMnO_4 + 5\,Pb(NO_3)_2 + 10\,HNO_3 + 4\,H_2O$

Q.4 (B)

Assume we have 1 mole sample which contains X_A mole of solute.

$\therefore$ Weight of solvent in 1 mole solution = $(1 - X_A)$. 17

$\because$ $17\,(1 - X_A)$ gm solvent contain X_A mole solute

$\therefore$ 1000 gm solvent contain $\frac{X_A \times 1000}{17(1-X_A)}$ mole solute = $\frac{58.8 \times A}{(1-X_A)}$

Q.5 (A)

$SO_2Cl_2 + 2H_2O \longrightarrow H_2SO_4 + 2HCl$

10	10	20

Number of moles of $SO_2Cl_2 = 50 \times 0.2 = 10$ millimoles

$\therefore$ Number of millimoles of $H_2SO_4 = 10$

$\therefore$ Number of millimoles of $HCl = 20$

$\therefore$ Number of millimoles of $H^+ = 20 + 20 = 40$

$\therefore$ Number of moles of $Ba(OH)_2 = 20$ millimoles

Now $V = \frac{20}{0.2} = 100$ ml

Q.6 (C)

98 gram		80 gram		
H_2SO_4	+	$2NaOH$	$\longrightarrow$	$Na_2SO_4 + 2H_2O$
H_3PO_4	+	$NaOH$	$\longrightarrow$	$Na_2HPO_4 + H_2O$
98 gram		40 gram		

40 gram NaOH reacts with 49 gram H_2SO_4 & 98 gram H_3PO_4 to form sulphate & dihydrogen phosphate.

wt. ratio

H_2SO_4 : H_3PO_4

49 : 98

1 :2 Ans.

Q.7 (B)

Let the wt. of sample = x g

Mass of silica = $x \times \frac{50}{100} = 0.5\,x$

Mass of water = $x \times \frac{15}{100} = 0.15\,x$

Mass of water left = $(0.15x - 10)$ g

New mass of sample = $(x - 10)$g

$$\%\text{ of silica} = \frac{0.5x}{(x-10)} \times 100 = 52 \qquad x = 260\text{g}$$

$$\%\text{ of water} = \frac{0.5x - 10}{(260-10)} \times 100 = 11.6\%$$

Alternative :

Let the mass of sample = 100 g

Mass of silica = 50 g, Mass of water = 15 g

Mass of all other components except water = 100 – 15 = 85 g

Since, on heating only water is removed hence composition of all other components will remain same

$$\frac{50}{85} = \frac{52}{(100-x)} \qquad (x \Rightarrow \%\text{ of water in final sample})$$

$x = 11.6$

Q.8 (C)

$B_2[H_2PtCl_6]_2 \longrightarrow Pt$

9 gm w gm

POAC for Pt :

$$\frac{9 \times 2}{2 \times 175 + 410 \times 2} = \frac{w}{195}$$

or, w = 3 gm Ans.

Q.9 (A)

Q.10 (A)

$10 \times 20 = 200$

$4 \times 480 = 1920$

$\Rightarrow$ 2120 gm Ans.

Q.11 (C)

Q.12 (B)

Q.13 (A)

(i)

$$\underbrace{\overset{a\,ml}{C_xH_yN_z} + \overset{(90-a)ml}{O_2}}_{90\,ml} \rightarrow \underset{40ml}{CO_2} + \underset{60ml}{H_2O} + \underset{20ml}{N_2}$$

Applying POAC for C

$a \times x = 40$ ...(i)

for H $y \times a = 120$ ...(ii)

for N $z \times a = 2 \times 20$ ...(iii)

for O $2 \times (90-a) = 2 \times 40 + 60$

$90 - a = 70$

$a = 20$ ml

(ii) Volume of $O_2 = 70$ ml

Mole % of O_2 = volume % of $O_2 = \frac{70}{90} \times 100 = \frac{700}{9}\%$

(iii) Solving (i), (ii), (iii)

$x = 2$

$y = 6$

$z = 2$

$C_2H_6N_2$

Q.14 (A) (B) (C) (D)

Using phenolphthalein,

$1 \times M_{Na_2CO_3} \times 10 = 2.5 \times \frac{1}{5} = 0.5$

$M_{Na_2CO_3} = 0.05$ M

Using methyl orange

$2 \times M_{Na_2CO_3} \times 10 + 1 \times M_{NaHCO_3} \times 10 = 7.5 \times \frac{1}{5}$

$2 \times 0.05 \times 10 + 10\,M_{NaHCO_3} = 1.5$

$M_{NaHCO_3} = 0.05$ M

Mass of $M_{Na_2CO_3}$ in 10 ml solution $= 10 \times 0.05 \times 106$ gm $\times 10^{-3} = 0.053$ gm

Mass of $NaHCO_3$ in 10 ml solution $= 10 \times 0.05 \times 84 \times 10^{-3}$ gm $= 0.042$ gm

Q.15 (A) S, (B) R, (C) Q, (D) P

%Mole

(A) % of (a – 1) = $\dfrac{a-(a+4)}{(a+4)-(a-1)} \times 100 = \dfrac{4}{5} \times 100 = 80\%$

(B) % of a = $\dfrac{(2a-5a)}{(5a-a)} \times 100 = \dfrac{3a}{4a} \times 100 = 75\%$

(C) % of (a + 1) = $\dfrac{(a+2)-(a+3)}{(a+3)-(a+1)} \times 100 = \dfrac{1}{2} \times 100 = 50\%$

(D) % of (a – 1) = $\dfrac{a-(a+2)}{(a+2)-(a-1)} \times 100 = \dfrac{2}{3} \times 100 = 66.67\ \%$

%Mass

(A) $\dfrac{80 \times (a-1)}{a} = \dfrac{80a-80}{a}$ (B) $\dfrac{75 \times a}{2a} = \dfrac{75a}{2a}$

(C) $\dfrac{50 \times (a+1)}{a+2} = \dfrac{5a+50}{a+2}$ (D) $\dfrac{66.67 \times (a-1)}{a} = \dfrac{66.67a-66.67}{a}$

Q.16 0.66

$X_{CH_3OH} = 0.1$

$\dfrac{40}{9}$ moles in 1000 gm ; $\dfrac{X}{\dfrac{40}{9}+\dfrac{1000}{18}+x} = 0.1$

$x = 0.1\left[\dfrac{40}{9}+\dfrac{1000}{18}+x\right]$; $10x = x + \dfrac{40}{9} + \dfrac{1000}{18}$

$9x = \dfrac{80+1000}{18}$; $x = \dfrac{1080}{18 \times 9}$

$\dfrac{n_{C_2H_5OH}}{n_{CH_3OH}} = \dfrac{40}{9} \times \dfrac{18 \times 9}{1080} = 0.66$ Ans.

Q.17 10 ml

Maximum volume is required in case of H_3PO_4

$H_3PO_4 \quad + \quad 3KOH \longrightarrow K_3PO_4 + 3H_2O$

$\dfrac{1.96}{98} = 0.2 \qquad 0.06$

$56 \times 0.06 = V \times \dfrac{33.6}{100}$

V = 10 ml Ans.

Q.18 6.8

$$N_2 + 3H_2 \longrightarrow 2NH_3$$

t = 0 1 3

t = final $1-x$ $3-3x$ $2x$

total mol = $4-2x$

$3NH_3 + H_3PO_4 \longrightarrow$ Product

Mole of $H_3PO_4 = M \times V$

$\Rightarrow$ $2.5 \times 0.2 = 0.5$ mol

for complete neutralization mol of $NH_3 = 0.5 \times 3 = 1.5$ mol

so that $= 2x = 1.5$

$x = 0.75$ mol

It means total mole in I[st] reaction $= 4 - 2x$

$= 4 - 2 \times 0.75$

$= 4 - 1.5 = 2.5$ mol

$$\overline{M} = \frac{\text{total mass}}{\text{total mol}} = \frac{28+6}{2.5} = 13.6$$

$$\text{V.D.} = \frac{\overline{M}\text{ wt.}}{2} = \frac{13.6}{2} = 6.8$$

Q.19 (a) 0.836, (b) 103.103

(a) Molality of $SO_3 = 2$

1000 gm of H_2SO_4 contains = 2 mole of SO_3

$$n_{H_2SO_4} = \frac{1000}{98}$$

$$n_{SO_3} = 2 \quad ; \quad X_{H_2SO_4} = \frac{\frac{1000}{98}}{\frac{1000}{98}+2} = 0.836$$

(b) 1160 gm mixture contains 2 mol SO_3

1160 gm can react with 36 gm H_2O

100 gm can react with $\frac{36}{1160} \times 100 = 3.103$

% labelling = 100 + 3.103 = 103.103

Q.20 12.5 *l*

Q.21 6 *l*

Q.22 1150

$$KMnO_4 + FeS_2 + CuS \longrightarrow Cu^{+2} + Fe^{+3} + SO_2$$

n= 11 n = 6

Eq. of $KMnO_4$ = Eq. of FeS_2 + Eq. of CuS

$$\frac{N \times V}{1000} = \frac{MV_{FeS_2}}{1000} \times n + \frac{MV_{CuS}}{1000} \times n$$

$$\frac{N \times 20}{1000} = \frac{10 \times 1}{1000} \times 11 + \frac{20 \times 1 \times 6}{1000}$$

$$N = \frac{110+120}{20} = \frac{230}{20} = 11.5\,N \text{ Ans.}$$

Q.23 90

$Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + I^{-1}$

$n_f = 1 \quad n_f = 2$

$0.1 \times 1 = 2 \times$ mole of I_2

$$\text{mole of } I_2 = \frac{0.1}{2} \text{ moles}$$

milli moles $= 0.1 \times 1000 / 2 = 50$

$IO_4^- + I_2 \longrightarrow IO_3^-$

$n_f = 2 \quad n_f = 10$

$10 \times$ mole of $I_2 = 2 \times 0.2 \times 1$

mole of $I_2 = 0.04$

milli mole $= 40$

total milli mole $= 40 + 50 = 90$

Q.24 0002

100 ml 11.2 V H_2O_2 (Molarity 1 M)

200 ml 2.8 m

$\rho = 1.0952$ gm/ml

$$\text{Molarity} = \frac{2.8}{1095.2/1.0952} \times 1000 = 2.8 \text{ M}$$

$$\text{Final molarity } M_1V_1 + M_2V_2 = M_f[V_1 + V_2] \times \frac{110}{100}$$

$$100 \times 1 + 2.8 \times 200 = M_f \times 300 \times \frac{110}{100}$$

$M_f = 2M$

Q.25 40

$$A^{n+} \longrightarrow A^{(n+4)+}$$

	$t = 0$	a	0	
T.R.	$t = t$	$a-x$	x	Let at both time 10 ml is titrated
	$t = 0,$	$a \times 10 \times 2 = 30 \times N$		Let Normality of titrating reagent = N

$$a = \frac{30}{10 \times 2} \times N = \frac{N}{10 \times 2} \times 30$$

At t = 10 min, $(a-x) \times 10 \times 2 + x \times 10 \times 5 = 45 \times N$

or $a \times 10 \times 2 - x \times 10 \times 2 + x \times 10 \times 5 = 45$ N

or $30x = 15$ N

$$x = \frac{15}{30}N = \frac{1}{2}N$$

$$K = \frac{1}{t} \ln \frac{a}{(a-x)} = \frac{1}{10} \ln \frac{3/2N}{3/2N - 1/2N}$$

$$= \frac{1}{10} \ln \frac{3/2}{1} = \frac{1}{10} \times 0.4 = 0.04$$

Q.26 0040

Mole of octane taken $= \frac{10 \times 1000 \times 1.14}{114} = 100$

$$C_8H_{18} + \frac{25}{2} O_2 \longrightarrow 8CO_2 + 9H_2O$$

x mole 8x mole 9x mole

$$C_8H_{18} + \frac{17}{2} O_2 \longrightarrow 8CO + 9H_2O$$

$(100-x)$ mole $8(100-x)$ mole $9(100-x)$ mole

from question : $8x \times 44 + 8(100-x) \times 28 + [9x + 9(100-x)] \times 18 = 43.72 \times 10^3$

$\therefore \quad x = 40$

Q.27 0004

11.2L at 0°C and 1 atm of compound $\Rightarrow \frac{1}{2}$ mole 'X' is always ≥ 15.5 g

$\therefore$ In 1 mole, of compound of 'X' ≥ 31 g

$\therefore$ Atmoic weight of 'X' = 31

($\because$ in one mole of compound of any element there should always be 1 mole of atoms of that element)

Also 11.2 L of vapour of 'X' weighs $\Rightarrow$ 62g

(At 0°C, 1 atm) $\left(\frac{1}{2}\text{mole}\right)$

$\therefore$ Molecular weight of 'X' = 124

Atomicity of 'X' $= \frac{124}{31} = 4$

Q.28 0134

$(COOH)_n + AgNO_3 \longrightarrow$ Silver salt $\longrightarrow$ white residue

Amount of residue obtained $(W_2) = 54$ gm

No. of moles of silver salt used $(n_1) = 0.25$

$$\frac{W_2}{108 \times n} = n_1$$

$$\frac{54}{108 \times n} = 0.25 \qquad n = 2$$

$$m_{(COONa)_2} = 2 \times 12 + 4 \times 16 + 2 \times 23 = 0134 \text{ gm/mole}$$

Q.29 0.0623M

Q.30 1.5, 40 ppm, pH = 2.6989